

4

Chapter 5: Acids, Bases and Salts

Paper 4 – (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	5 – Acids, Bases and Salts
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic P4 traps: confusing the colour an indicator shows with the family it names (china rose magenta=acid not base, methyl orange red=acid, turmeric red=base, phenolphthalein pink=base); reading one indicator as conclusive when a second contradicts or merely fails to distinguish (phenolphthalein/turmeric staying unchanged cannot separate acid from neutral); treating 'no litmus change' as proof of water rather than mere neutrality; over- and under-shooting an endpoint during drop-by-drop neutralisation; choosing the wrong member of the acid-base pair to fix a sting, soil, effluent or stomach; assuming the salt from every neutralisation is common salt; and confusing strength or amount of taste with chemical family. Do not write on this paper — mark answers on your answer sheet.

1. A laboratory has four unlabelled colourless liquids known to be lemon juice, sugar water, lime water and soap solution, in some unknown order. A single strip of blue litmus is dipped into each and turns red in exactly one of them. That one liquid must be:
(A) lemon juice (B) lime water
(C) soap solution (D) sugar water
2. Among curd, milk of magnesia, lime water and vinegar, suppose you remove every substance that turns blue litmus red. The substances that remain are:
(A) curd and vinegar (B) vinegar and lime water
(C) milk of magnesia and lime water (D) only milk of magnesia
3. Tomato juice (an acid) and milk of magnesia (a base) are mixed in exactly neutralising amounts. Predicting what red litmus does with the mixture, and then what happens if one extra drop of tomato juice is added and blue litmus is tested, gives:

- (A) red litmus shows no change, then the extra drop turns blue litmus red
- (B) red litmus turns blue, then there is no further change
- (C) red litmus shows no change, then the extra drop turns red litmus blue
- (D) both stay unchanged forever, because neutralisation cancels everything permanently

4. Three clear liquids — distilled water, common salt water and sugar water — are each tested with red and blue litmus. A student says only distilled water will leave both papers unchanged. The most accurate evaluation is:

- (A) all three leave both litmus papers unchanged because all three are neutral, so litmus cannot single out the distilled water
- (B) only distilled water leaves both unchanged; salt water turns blue litmus red
- (C) only distilled water leaves both unchanged; sugar water turns red litmus blue
- (D) all three turn blue litmus red, because dissolved solids are acidic

5. Dry turmeric paper is touched in turn with soap solution, then lemon juice, then plain water. Recording the colours in that order gives:

- (A) yellow, red, yellow
- (B) red, red, yellow
- (C) red, yellow, yellow
- (D) yellow, yellow, red

6. A drop of china rose indicator is added to vinegar and separately to soap solution. Recalling that china rose is dark pink (magenta) in acids and green in bases, the two colours seen are:

- (A) green in vinegar and magenta in soap solution
- (B) magenta in both, because china rose is always pink
- (C) magenta in vinegar and green in soap solution
- (D) colourless in vinegar and pink in soap solution, like phenolphthalein

7. Phenolphthalein is added to four liquids: tamarind water, lime water, sugar solution and milk of magnesia. The number of these that turn phenolphthalein pink is:

- (A) 1
- (B) 3
- (C) 2
- (D) 0

8. Methyl orange is red in acids and yellow in bases. A drop is added to a fresh sample of lime water, and a separate drop to dilute hydrochloric acid. The colours are, respectively:

- (A) yellow and red
- (B) red and yellow
- (C) pink and colourless
- (D) yellow and yellow

9. Milk of magnesia tastes only mildly bitter. A student argues that because it is merely mildly bitter, it must really be an acid rather than a base. The correct judgement is:

- (A) correct; only strongly bitter things are bases
- (B) correct; mild bitterness means it is an acid

(C) wrong; mild bitterness means it is neutral

(D) wrong; a bitter taste, even a mild one, is a base property, and its intensity does not change the family

10. Hydrochloric acid is neutralised once by sodium hydroxide and once by potassium hydroxide. After each mixture is evaporated, the solids left behind are:

(A) common salt (sodium chloride) in both cases, because the acid was the same

(B) potassium chloride in both cases, because potassium is the stronger metal

(C) no solid in either case, because neutralisation destroys all matter

(D) two different salts — sodium chloride from the first and potassium chloride from the second — because the bases differ

11. A pink, phenolphthalein-treated washing soda solution is titrated by adding vinegar drop by drop. The pink has just vanished; the student then adds several more drops of vinegar. With phenolphthalein still in the mixture, the colour is now:

(A) pink again, because adding more liquid restores the colour

(B) pink again, because excess acid turns phenolphthalein pink

(C) deep red, because excess acid reddens phenolphthalein

(D) still colourless, because the excess vinegar keeps the mixture non-basic

12. Four liquids are tested with litmus: P turns red litmus blue, Q turns blue litmus red, R changes neither litmus, and S turns blue litmus red. Turmeric paper is then touched to each. The number that stay yellow is:

(A) 1

(B) 3

(C) 2

(D) 4

13. When sulphuric acid reacts with sodium hydroxide the salt is sodium sulphate — named from the sodium of the base and the sulphate of the acid. By the same naming rule, the salt formed from hydrochloric acid and potassium hydroxide is:

(A) common salt (sodium chloride)

(B) sodium chloride again, because chloride always pairs with sodium

(C) potassium chloride

(D) hydrochloric acid, regenerated by the reaction

14. A pinch of baking soda is stirred into lemon juice and fizzes; the same pinch stirred into sugar water gives almost no fizz. The best explanation of the difference is:

(A) lemon juice is a base and sugar water is an acid, so only lemon juice reacts

(B) baking soda is an acid that fizzes only with the sugar water

(C) baking soda (a base) reacts with the acid in lemon juice to release a gas, but sugar water is neutral so there is little reaction to release gas

(D) the fizz is just trapped air leaving the baking soda, the same in both liquids

15. A child brushes against stinging nettle (which injects an acid) and is later stung by a wasp (whose sting is alkaline/basic). The correct single-line first-aid for the two,

respectively, is:

- (A) baking soda paste for both stings
- (B) mild vinegar for the nettle sting and baking soda paste for the wasp sting
- (C) baking soda paste for the nettle sting and a mild acid such as vinegar for the wasp sting
- (D) lemon juice for the nettle sting and vinegar for the wasp sting

16. Litmus comes from lichens, turmeric from a root, and china rose (gudhal) from a flower. By contrast, phenolphthalein is best described as:

- (A) an indicator extracted from red cabbage
- (B) a synthetic chemical made in the laboratory, not obtained from any plant or living source
- (C) an indicator obtained from the bark of a tree
- (D) a natural indicator just like the other three

17. A farmer adds quicklime (a base) to a field that is too acidic, and the crop improves. He then spreads the same quicklime on a neighbouring field that is already slightly basic. The likely result on the second field is:

- (A) it becomes even more basic and worse for the crop, because adding a base to basic soil increases its basicity
- (B) it becomes neutral, because lime balances any soil
- (C) it becomes acidic, because lime is an acid
- (D) no change, because lime is an inert salt

18. Phenolphthalein is added to a beaker of dilute acid and stays colourless. Sodium hydroxide is then added drop by drop, with stirring, until the liquid just turns pink. At that moment the beaker has become:

- (A) still acidic, because acids are stronger than the base
- (B) exactly neutral, since pink marks the neutral point
- (C) a fresh stronger acid, because mixing two liquids strengthens the acid
- (D) basic, since enough base was added to pass neutral and turn phenolphthalein pink

19. A liquid is reported to turn phenolphthalein pink yet leave both blue and red litmus papers completely unchanged. The most reasonable conclusion is:

- (A) the liquid is a special neutral substance, since litmus did not change
- (B) the liquid is an acid, because phenolphthalein went pink
- (C) the liquid is salt water, which behaves this way
- (D) the report is inconsistent, because anything basic enough to turn phenolphthalein pink would also turn red litmus blue

20. Solution X turns methyl orange yellow; solution Y turns methyl orange red; solution Z leaves both litmus papers unchanged. Recalling methyl orange is red in acids and yellow in bases, the families of X, Y and Z are:

(A) acid, base, neutral

(B) base, neutral, acid

(C) base, acid, neutral

(D) acid, neutral, base

21. An antacid tablet contains magnesium hydroxide, a mild base. When it reaches a stomach holding excess hydrochloric acid, the chemistry that gives relief is best described as:

(A) the base adds to the acid, making the stomach more acidic and easing the burn

(B) the tablet coats the stomach so the acid simply cannot be felt, with no reaction

(C) the base turns the stomach acid into a stronger acid that the body then expels

(D) the mild base neutralises the excess acid, forming a salt and water, so the burning eases

22. A liquid turns red litmus blue. Predicting what blue litmus, phenolphthalein and turmeric will each do with the same liquid gives:

(A) blue litmus stays blue, phenolphthalein turns pink, turmeric turns red

(B) blue litmus turns red, phenolphthalein stays colourless, turmeric stays yellow

(C) blue litmus stays blue, phenolphthalein stays colourless, turmeric stays yellow

(D) blue litmus turns red, phenolphthalein turns pink, turmeric turns red

23. Soda water contains dissolved carbon dioxide, which forms carbonic acid. Predicting how it affects blue litmus and methyl orange gives:

(A) blue litmus turns red and methyl orange turns yellow

(B) neither changes, because the drink is neutral

(C) blue litmus stays blue and methyl orange turns red

(D) blue litmus turns red and methyl orange turns red

24. Old marble (calcium carbonate) buildings in cities develop pitted, eaten-away surfaces faster than the same marble in clean countryside air. The best chemical reason is:

(A) city rain is basic and reacts with the acidic marble, dissolving it

(B) city air carries more sulphur dioxide and nitrogen oxides, making the rain more acidic, and that acid reacts with the basic calcium carbonate, eating the marble away

(C) city rain stays neutral and merely washes loose soot from the surface

(D) marble in cities slowly turns into an acid and reacts with itself

25. Onion extract and vanilla essence are olfactory indicators: their smell stays in an acid but fades or disappears in a basic solution. A cloth rubbed with onion extract loses its oniony smell when dipped in a certain solution. That solution is most likely:

(A) an acid

(B) a base

(C) pure water

(D) common salt water

26. A drop of china rose indicator turns cup A magenta and cup B green. Methyl orange is then added to fresh portions of A and B. Recalling methyl orange is red in acids and yellow in bases, the colours seen are:

(A) yellow in A and red in B

(B) red in A and yellow in B

(C) red in both A and B

(D) yellow in both A and B

27. Vinegar is added drop by drop to a beaker of soap solution that contains phenolphthalein and is currently pink. After many drops the pink just disappears; the experimenter keeps adding vinegar. Testing the final mixture with blue litmus would now show:

(A) no change, because the mixture is locked at neutral

(B) red litmus turns blue, because soap solution is always basic

(C) blue litmus turns red, because the excess vinegar has made the mixture acidic

(D) blue litmus turns green, a sign of overshoot

28. Small marble chips (calcium carbonate) are dropped into two beakers: one of dilute hydrochloric acid, one of lime water. Predicting where bubbling (gas release) is seen gives:

(A) bubbling in the lime-water beaker only, where the base reacts with the marble

(B) bubbling in the hydrochloric-acid beaker only, where the carbonate reacts with the acid to release gas

(C) bubbling in both beakers equally

(D) no bubbling in either, because marble is unreactive

29. A workshop's acidic waste water must be neutralised before release into a river. An engineer adds slaked lime until litmus shows neutral, but a colleague urges adding more lime 'to be safe'. The best advice is:

(A) keep adding lime, because more base always makes water safer

(B) once neutral, add some acid instead to be sure

(C) add common salt to lock in the neutral state

(D) stop at neutral; adding excess lime would make the water basic and also harm river life

30. A liquid turns methyl orange red, which already shows it is an acid. Predicting its effect on blue litmus and on phenolphthalein then gives:

(A) it is an acid: blue litmus turns red and phenolphthalein stays colourless

(B) it is a base: blue litmus stays blue and phenolphthalein turns pink

(C) it is neutral: neither litmus changes and phenolphthalein stays colourless

(D) it is an acid: blue litmus stays blue and phenolphthalein turns pink

31. Among vinegar, soap solution, ammonia solution, lemon juice and sugar water, the number that are bases is:

(A) 2

(B) 1

(C) 3

(D) 4

32. A label-less liquid turns red litmus blue and turns methyl orange yellow, but the tester refuses to taste it. The safest single classification is:

- (A) an acid, because methyl orange changed colour
- (B) neutral, because two different tests can never agree
- (C) a base, since red litmus turning blue and methyl orange turning yellow both point the same way
- (D) impossible to classify without tasting it

33. Four substances are to be sorted: lemon juice, ammonia solution, sugar water and washing soda. Grouped as (acids) / (bases) / (neutral), the correct grouping is:

- (A) lemon juice and sugar water / ammonia solution / washing soda
- (B) ammonia solution / lemon juice and washing soda / sugar water
- (C) washing soda / lemon juice / ammonia solution and sugar water
- (D) lemon juice / ammonia solution and washing soda / sugar water

34. A child is stung by a bee (an acidic sting) and rubs vinegar on it, thinking any sharp 'medicine-like' liquid will help. The best assessment is:

- (A) correct; vinegar relieves every kind of sting
- (B) correct; vinegar is a base that neutralises the acidic bee sting
- (C) the bee sting is basic, so vinegar is the right acid for it
- (D) wrong; the bee sting is acidic, so vinegar (an acid) will not neutralise it — a mild base such as baking soda paste is needed

35. A liquid leaves phenolphthalein colourless. A student insists this proves the liquid is neutral. The most accurate correction is:

- (A) correct; colourless phenolphthalein always means neutral
- (B) wrong; colourless phenolphthalein always means a base
- (C) wrong; colourless phenolphthalein always means an acid
- (D) colourless phenolphthalein only shows the liquid is not a base; it could be acidic or neutral, so a litmus test is needed to decide

36. A student must tell pure water apart from a very weak acid using a single indicator. The most reliable test, and why, is:

- (A) phenolphthalein — water turns it faintly pink while the acid does not
- (B) litmus — water leaves both papers unchanged, while the weak acid turns blue litmus red
- (C) turmeric — water turns it red while the acid does not
- (D) taste — water tastes sweet while the acid tastes salty

37. Ammonia solution (a base) is mixed bit by bit with vinegar until red litmus stays red and blue litmus also stays blue. At that point the mixture is:

- (A) still basic, because the ammonia can never be fully cancelled
- (B) neutral, because just enough vinegar has neutralised the ammonia so neither litmus changes

- (C) now acidic, because any added acid makes a mixture acidic
- (D) a fresh base stronger than ammonia

38. Concentrated lemon juice and the same lemon juice after being diluted with plenty of water are each tested with blue litmus. The results are:

- (A) both turn blue litmus red, because both are acidic; dilution lowers the strength but not the family
- (B) only the concentrated juice turns blue litmus red; the dilute one leaves it unchanged
- (C) the dilute juice keeps blue litmus blue while the concentrated one turns it red
- (D) neither changes litmus, because lemon juice becomes neutral once water is added

39. Washing soda solution turns red litmus blue. Tracing what the same solution does to turmeric paper and to phenolphthalein gives:

- (A) turmeric stays yellow and phenolphthalein stays colourless
- (B) turmeric turns red and phenolphthalein turns pink
- (C) turmeric turns red and phenolphthalein stays colourless
- (D) turmeric stays yellow and phenolphthalein turns pink

40. To protect a limestone wall (calcium carbonate) damaged by acid rain, two treatments are proposed: (i) periodically washing it with a very dilute basic solution, or (ii) washing it with a dilute acid. The chemically sound choice, and reason, is:

- (A) the acid wash, because only acid can clean acid off stone
- (B) the basic wash, because a base neutralises the acidic residue left by acid rain on the stone
- (C) neither, because limestone is acidic and unaffected by any wash
- (D) both work equally, since any wash simply removes residue

41. Orange juice, soap solution and sugar water are each tested with blue litmus, then with phenolphthalein. The substance that keeps blue litmus blue AND turns phenolphthalein pink is:

- | | |
|-------------------|------------------|
| (A) orange juice | (B) sugar water |
| (C) soap solution | (D) none of them |

42. Dilute hydrochloric acid is added little by little to a beaker of lime water until neither litmus paper changes. A few more drops of the acid are then added. The mixture is now:

- (A) still exactly neutral, because the reaction is finished
- (B) slightly acidic, because the extra acid is now in excess after neutralisation
- (C) basic, because lime water is always the stronger one
- (D) unchanged, because acid cannot affect a neutral mixture

43. Sulphuric acid is neutralised by potassium hydroxide, and a student labels the salt formed 'common salt'. The correct label, with reason, is:

(A) common salt (sodium chloride), as in all neutralisations

(B) no salt forms, only water

(C) sulphuric acid, regenerated by the reaction

(D) potassium sulphate, since the salt comes from this particular acid-base pair, not sodium chloride

44. Why can turmeric paper alone fail to tell an acid apart from a neutral liquid, so that a second indicator is needed?

(A) turmeric stays yellow in both acids and neutral liquids — it turns red only in bases — so a yellow result cannot separate an acid from a neutral liquid

(B) turmeric turns red in both acids and neutral liquids, giving the same result for each

(C) turmeric turns the same colour in acids and in bases, so it cannot tell them apart

(D) turmeric changes colour only in neutral liquids, never in acids or bases

45. An indicator chart must list one indicator made in a laboratory beside one taken from nature. The correctly mixed pair (one synthetic, one natural) is:

(A) methyl orange (synthetic) and china rose (natural)

(B) litmus (synthetic) and turmeric (natural)

(C) phenolphthalein (natural) and litmus (synthetic)

(D) china rose (synthetic) and turmeric (natural)

46. While neutralising dilute hydrochloric acid with sodium hydroxide, a student feels the beaker warm up and wonders whether the acid is 'boiling away'. The correct explanation is:

(A) no; the warmth is heat released by the neutralisation reaction itself, not the acid boiling away

(B) yes; the acid is boiling away, which always heats the beaker

(C) no; the warmth means the base is freezing the mixture

(D) yes; the salt being formed boils off and heats the glass

47. A lake has turned acidic and fish are dying. Of the following bases, the cheap, bulk option actually used to treat such water, rather than an expensive laboratory chemical, is:

(A) sodium hydroxide (caustic soda) (B) ammonia solution

(C) slaked lime (calcium hydroxide) (D) milk of magnesia

48. Three liquids X, Y, Z are tested. X turns blue litmus red. Y turns red litmus blue. Z changes neither litmus. Which single liquid, treated with china rose indicator (magenta in acids, green in bases), turns it magenta?

(A) X

(B) Y

(C) Z

(D) both X and Y

49. A student tests an unknown liquid: it turns blue litmus red, turns methyl orange red and leaves phenolphthalein colourless. The single best conclusion is:

- (A) it is a base, because methyl orange changed colour
- (B) it is neutral, because phenolphthalein stayed colourless
- (C) it is an acid, since blue-litmus-to-red and methyl-orange-red both indicate acidity, and colourless phenolphthalein is consistent with that
- (D) the results conflict, so no conclusion is possible

50. A solution turns china rose green. Without doing any further test, which single prediction about this solution is correct?

- (A) it will taste sour
- (B) it will turn red litmus blue
- (C) it will turn blue litmus red
- (D) it will leave phenolphthalein colourless

51. Of the following everyday events, the one that is genuinely an 'acid + base $\rightarrow$ salt + water' neutralisation is:

- (A) dissolving sugar in tea
- (B) ice melting in a glass of juice
- (C) a slice of apple turning brown in air
- (D) brushing the teeth with toothpaste (a mild base) to neutralise the acid made by mouth bacteria

52. Lemon juice is exactly neutralised by baking soda solution, and the water is then evaporated off. A student expects to recover the original lemon juice. Instead, what is left behind is:

- (A) the original lemon juice, simply dried out
- (B) leftover baking soda, which never reacts fully
- (C) a salt formed in the neutralisation, since the acid was used up and no acid remains
- (D) nothing, because neutralisation removes all matter

53. A chart reads: lemon juice — acid, milk of magnesia — base, vinegar — acid, common salt — base, soap solution — base. Exactly one pairing is wrong. It is:

- (A) lemon juice — acid
- (B) common salt — base
- (C) milk of magnesia — base
- (D) soap solution — base

54. A liquid turns blue litmus red, but a classmate worries the litmus strip may have been faulty. Which independent test result would best back up that the liquid really is an acid?

- (A) methyl orange turns red in it
- (B) phenolphthalein turns pink in it
- (C) turmeric paper turns red in it
- (D) red litmus turns blue in it

55. A student finds that tap water leaves both litmus papers unchanged and declares it 'chemically pure water'. Why is this unsafe to conclude?

- (A) many neutral liquids — salt water, sugar water and the like — also leave litmus unchanged, so the test shows only neutrality, not that it is pure water
- (B) pure water actually turns blue litmus red, so no change means it is not water
- (C) only acids and bases leave litmus unchanged, never water
- (D) litmus cannot be used on any liquid that is colourless

56. When dilute hydrochloric acid is neutralised by sodium hydroxide solution, the equation is $\text{acid} + \text{base} \rightarrow \text{salt} + \text{water}$. A student asks where the water in the products comes from. The correct identification is:

- (A) it is only the water the chemicals were already dissolved in, and no new water forms
- (B) it is leftover hydrochloric acid
- (C) there is no water among the products, only salt
- (D) it is genuinely a new product formed when the acid and base react, alongside the salt

57. A liquid turns blue litmus red. Asked whether it could be soap solution, distilled water, vinegar or milk of magnesia, the only one that fits is:

- (A) soap solution
- (B) distilled water
- (C) vinegar
- (D) milk of magnesia

58. A student tests four liquids with blue litmus: lemon juice turns it red, soap solution leaves it unchanged, vinegar turns it red, lime water leaves it unchanged. The number of acids identified is:

- (A) 1
- (B) 3
- (C) 4
- (D) 2

59. Cup P turns china rose magenta and cup Q turns china rose green. To neutralise cup Q you would slowly add cup P. This works because cup P is:

- (A) a base, because china rose magenta marks a base
- (B) the acid, because china rose magenta marks an acid, and an acid neutralises the base in cup Q
- (C) neutral, so it merely dilutes cup Q without neutralising it
- (D) the same as cup Q, since china rose cannot tell acids from bases

60. At the endpoint of a neutralisation, a student wants one indicator result that confirms neither acid nor base is in excess. Why is phenolphthalein a poor choice for spotting the exact neutral point, and what is better?

- (A) phenolphthalein turns red exactly at neutral, which is too faint to see; turmeric is better
- (B) phenolphthalein is colourless across the neutral range and the whole acidic range, so it does not flag the exact neutral point; two unchanged litmus papers (blue stays blue, red stays red) are a better check
- (C) phenolphthalein works only in solutions free of any base; methyl orange is better
- (D) phenolphthalein cannot be used in any solution that contains water; taste is better

Solutions — Science (Class 7), Chapter 5: Acids, Bases and Salts — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	D	21	D	31	A	41	C	51	D
2	C	12	B	22	A	32	C	42	B	52	C
3	A	13	C	23	D	33	D	43	D	53	B
4	A	14	C	24	B	34	D	44	A	54	A
5	C	15	C	25	B	35	D	45	A	55	A
6	C	16	B	26	B	36	B	46	A	56	D
7	C	17	A	27	C	37	B	47	C	57	C
8	A	18	D	28	B	38	A	48	A	58	D
9	D	19	D	29	D	39	B	49	C	59	B
10	D	20	C	30	A	40	B	50	B	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Blue litmus turns red only in an acid. Of the four, only lemon juice (citric acid) is an acid, so it is the single liquid that turns blue litmus red. Lime water and soap solution are both bases, which leave blue litmus unchanged (they turn red litmus blue instead). Sugar water is neutral and leaves blue litmus unchanged. Since exactly one strip turned red, that liquid is the acid, i.e. lemon juice.

2. (C) Turning blue litmus red is an acid property, so removing every such substance removes all the acids. Curd (lactic acid) and vinegar (acetic acid) are acids, so they are removed. Milk of magnesia (magnesium hydroxide) and lime water are bases, which do not turn blue litmus red, so both remain. Keeping only milk of magnesia would wrongly discard the equally basic lime water; keeping vinegar would keep an acid.

3. (A) Tomato juice is an acid and milk of magnesia is a base. Mixed in exactly neutralising amounts they cancel, leaving the mixture neutral, so red litmus shows no change (it is already red, and only a base would turn it blue). Adding one extra drop of

tomato juice tips the mixture slightly acidic, so it now turns blue litmus red. It would not turn red litmus blue (that needs excess base), and the neutral state is not permanent once more acid is added.

4. (A) Leaving both litmus papers unchanged is exactly the result of any neutral liquid. Distilled water, common salt water and sugar water are all neutral, so all three leave red and blue litmus unchanged and give identical results. The test therefore establishes neutrality, not identity; it cannot single out the distilled water. Salt water does not turn blue litmus red and sugar water does not turn red litmus blue (both are neutral, not acidic or basic), and dissolved solids do not make these solutions acidic.

5. (C) Turmeric turns red only in a base and stays yellow in acids and in neutral liquids. Soap solution is basic, so the paper turns red. Lemon juice is acidic, so the paper stays yellow. Plain water is neutral, so the paper stays yellow. The order is therefore red, yellow, yellow. Reversing red to the acid (lemon) misremembers turmeric's direction; turmeric does not turn red in acids or in water.

6. (C) China rose shows dark pink (magenta) in acids and green in bases. Vinegar is acidic (acetic acid), so it turns china rose magenta. Soap solution is basic, so it turns china rose green. Swapping the two reverses china rose's direction. China rose is not a single fixed colour, and it is not colourless like phenolphthalein in acid; it goes magenta there.

7. (C) Phenolphthalein turns pink only in a base. Lime water and milk of magnesia are bases, so both turn it pink, giving a count of 2. Tamarind water is acidic (tartaric acid) and sugar solution is neutral, so both leave phenolphthalein colourless. Counting 1 forgets that both lime water and milk of magnesia are bases; counting 3 wrongly includes an acid or neutral.

8. (A) Methyl orange is yellow in bases and red in acids. Lime water is a base, so methyl orange is yellow in it; dilute hydrochloric acid is an acid, so methyl orange is red in it. The order is yellow then red. Reversing them confuses the directions; methyl orange shows neither pink nor colourless (those are phenolphthalein behaviours), and the acid would not also be yellow.

9. (D) Bitter taste is a signature of a base no matter how strong or weak it is, and milk of magnesia (magnesium hydroxide) is a base. A mild bitterness simply means a weakly basic taste, not an acid; the intensity of a taste does not switch the chemical family. An acid would taste sour, not bitter, so mild bitterness cannot point to an acid, and a neutral liquid has no bitter taste at all.

10. (D) Neutralisation gives a salt plus water, and the salt depends on which acid and base reacted. Here the acid (hydrochloric acid) is the same both times, but the bases differ: sodium hydroxide gives sodium chloride, while potassium hydroxide gives potassium chloride. So the two solids are different salts. They are not both common salt or both

potassium chloride, none of the base remains after exact neutralisation, and matter is not destroyed, so a solid salt does form each time.

11. (D) Phenolphthalein is pink only in a base and colourless otherwise. While the washing soda solution was basic the indicator was pink; when the pink vanished the base had been neutralised and the solution was about neutral. Adding more vinegar (acid) makes the solution acidic, which is also colourless for phenolphthalein, so it stays colourless. Adding more liquid does not bring back pink, excess acid never turns phenolphthalein pink, and phenolphthalein has no deep-red colour.

12. (B) Turmeric turns red only in a base and stays yellow in acids and neutral liquids. P turns red litmus blue, so P is a base and turns turmeric red. Q turns blue litmus red, so Q is an acid and leaves turmeric yellow. R changes neither litmus, so R is neutral and leaves turmeric yellow. S turns blue litmus red, so S is an acid and leaves turmeric yellow. Hence three of the four (Q, R and S) stay yellow, and only P (the base) turns red.

13. (C) A salt is named from the metal of the base combined with the acid part of the acid, exactly as the worked example shows (sodium from the base + sulphate from the acid → sodium sulphate). Applying the same rule, hydrochloric acid (acid part: chloride) with potassium hydroxide (metal: potassium) gives potassium chloride. It is not sodium chloride, since no sodium is present — chloride does not 'always' pair with sodium — and the acid is not regenerated by neutralisation.

14. (C) Baking soda is a base; lemon juice is an acid, so they undergo a neutralisation that releases carbon dioxide gas, seen as fizzing. Sugar water is neutral, so there is no acid for the baking soda to react with and little gas is produced. Lemon juice is the acid (not a base) and sugar water is neutral (not an acid). The bubbles are a gas produced by a real reaction, not trapped air, which is why they appear strongly only with the acidic lemon juice.

15. (C) Each sting is relieved by neutralising it with the opposite family. Stinging nettle injects an acid, so a mild base — baking soda paste — neutralises it. The wasp sting is basic, so a mild acid such as vinegar neutralises it. Using baking soda on both would worsen the basic wasp sting, and lemon juice (an acid) is useless on the acidic nettle sting. Only baking soda for the nettle and vinegar for the wasp pairs each sting with the opposite family.

16. (B) Phenolphthalein is a synthetic, lab-made indicator, unlike the three named natural ones. Litmus comes from lichens, turmeric from the turmeric root and china rose from hibiscus flowers, so all three are obtained from plants or living sources. Phenolphthalein is not extracted from red cabbage or tree bark (those would be natural sources) and is not a natural indicator; it is manufactured chemically.

17. (A) Quicklime is a base. Added to acidic soil it neutralises the excess acid and improves the field, which is why the first crop improves. Spread on a field that is already

basic, it adds more base and makes the soil even more basic, worsening it. Lime is not neutral, so it does not balance any soil; it is not an acid, so it does not lower basicity; and it is not an inert salt, since it clearly changes acidic soil.

18. (D) Phenolphthalein is colourless in acids and turns pink only in a base. Starting colourless, the beaker held an acid. Adding sodium hydroxide neutralises that acid, and the moment the liquid turns pink shows enough base has been added to pass the neutral point and make the mixture basic. So the beaker is now basic, not still acidic. Pink does not mark the neutral point (a neutral mixture leaves phenolphthalein colourless), and adding a base neutralises an acid rather than strengthening it.

19. (D) Phenolphthalein turns pink only in a base, and a base also turns red litmus blue. A liquid cannot be basic for phenolphthalein yet neutral for litmus at the same time, so the two results contradict each other and the report is inconsistent. A neutral substance leaves phenolphthalein colourless, an acid leaves phenolphthalein colourless too (pink means base, not acid), and salt water is neutral, so none of those explain a genuine pink-with-no-litmus-change result.

20. (C) Methyl orange is yellow in bases and red in acids. Solution X turns it yellow, so X is a base. Solution Y turns it red, so Y is an acid. Solution Z changes neither litmus, so Z is neutral. Hence base, acid, neutral. The other orders mislabel the methyl-orange-yellow solution as an acid or treat the methyl-orange-red solution as something other than an acid.

21. (D) Acidity is too much hydrochloric acid in the stomach. Magnesium hydroxide is a mild base, and a base neutralises an acid to give a salt and water, removing the excess acid and easing the burning. A base does not make the stomach more acidic; that would worsen the problem. The relief is chemical neutralisation, not a coating that merely hides the acid, and an antacid does not strengthen the acid.

22. (A) Red litmus turning blue marks a base. In a base, blue litmus shows no change (it is already blue and only an acid turns it red), phenolphthalein turns pink (it is pink only in a base), and turmeric turns red (it turns red only in a base). So the pattern is blue litmus unchanged, phenolphthalein pink, turmeric red. The other options apply acid behaviours (blue litmus to red, phenolphthalein colourless, turmeric yellow) to what is actually a base.

23. (D) Dissolved carbon dioxide makes carbonic acid, so soda water is mildly acidic. An acid turns blue litmus red, and methyl orange is red in acids, so both of those happen. The drink is not basic, so methyl orange would not turn yellow and blue litmus would not stay blue. It is not neutral, so litmus does change. Only 'blue litmus red and methyl orange red' fits an acid on both counts.

24. (B) City air carries more sulphur dioxide and nitrogen oxides, which dissolve in rain to form acids, so city rain is more acidic. Marble is calcium carbonate, which behaves as a

base, so the acidic rain reacts with and gradually eats away the basic marble — faster in polluted city air than in clean countryside air. The rain is not basic and the marble is not acidic, so the reversed chain is wrong; the rain is acidic, not neutral, and the marble does not react 'with itself'.

25. (B) Olfactory indicators such as onion and vanilla keep their smell in an acid but lose it in a basic solution. So a cloth that loses its oniony smell when dipped in a liquid has been dipped in a base. An acid would leave the onion smell intact, and pure water or common salt water are neutral, where the smell would also remain. The disappearance of the smell is the test result that marks a base.

26. (B) China rose is magenta in acids and green in bases, so cup A (magenta) is an acid and cup B (green) is a base. Methyl orange is red in acids and yellow in bases, so it turns red in A (acid) and yellow in B (base). Reversing this would treat magenta as basic, which is wrong. Both red would need both to be acids; both yellow would need both to be bases, but A is an acid.

27. (C) While pink, the soap solution was basic. The pink vanishing marks the point where the vinegar (acid) has neutralised the base, leaving the mixture about neutral. Continuing to add vinegar makes the mixture acidic, so blue litmus now turns red. The mixture is not locked at neutral once more acid is added; the soap solution has been used up so it is no longer basic; and litmus has no green result.

28. (B) Marble is calcium carbonate, which reacts with an acid to release carbon dioxide gas, seen as bubbling. The hydrochloric-acid beaker therefore bubbles. Lime water is basic, and a carbonate does not release gas with a base in this way, so the lime-water beaker does not bubble. Bubbling in both, or none at all, ignores that the reaction needs an acid. Marble is not unreactive towards acids.

29. (D) Once litmus shows the acidic waste has reached neutral, the acid has been used up, and the water is safe. Adding excess slaked lime (a base) would push the water past neutral into basic, which is itself harmful to river life, so the colleague's advice is wrong. Adding acid again or adding common salt does not improve safety either; the correct move is to stop at neutral. 'More base is always safer' is the trap, since over-treating swaps one harm for another.

30. (A) Methyl orange is red only in an acid, so a liquid that turns it red is an acid. In an acid, blue litmus turns red and phenolphthalein stays colourless (it is pink only in a base). So the liquid is an acid with blue litmus red and phenolphthalein colourless. The base options (blue litmus unchanged with phenolphthalein pink) and the neutral option contradict the methyl-orange-red result, which already fixes the liquid as an acid.

31. (A) Bases turn red litmus blue and feel soapy. Soap solution and ammonia solution are bases, giving a count of 2. Vinegar (acetic acid) and lemon juice (citric acid) are acids, and

sugar water is neutral, so none of those three is a base. A count of 1 misses one base, while 3 or 4 wrongly counts an acid or the neutral sugar water as basic.

32. (C) Red litmus turning blue is a base property, and methyl orange turning yellow is also a base property (it is yellow in bases, red in acids). The two results agree, so the liquid is a base. Methyl orange changing colour does not by itself mean acid — its yellow specifically marks a base. A neutral substance would not turn red litmus blue, so two agreeing base results rule out neutral, and tasting is unnecessary and unsafe.

33. (D) Lemon juice (citric acid) is the acid. Ammonia solution and washing soda solution are both bases. Sugar water is neutral. So acids = lemon juice, bases = ammonia solution and washing soda, neutral = sugar water. Sugar water is not an acid, ammonia is not the lone base, and washing soda is not neutral, so the other groupings misplace at least one substance.

34. (D) A bee sting is acidic and is relieved by neutralising it with the opposite family — a mild base such as baking soda paste. Vinegar is an acid, so rubbing it on an acidic sting adds the same family and does not neutralise it. Vinegar is not a base, the bee sting is acidic (not basic), and being 'sharp' or 'medicine-like' has nothing to do with neutralising power. The correct remedy is a mild base.

35. (D) Phenolphthalein is pink in bases and colourless in both acids and neutral solutions. So a colourless result only rules out a base; it cannot by itself tell an acid from a neutral substance. A second test, such as blue litmus, is needed: if it turns red the solution is an acid, otherwise neutral. Colourless does not 'always' mean neutral, base or acid, since it covers two of the three families at once.

36. (B) Litmus separates pure water from a weak acid in one step: water leaves both papers unchanged (it is neutral), while even a weak acid turns blue litmus red. Phenolphthalein is useless here because water leaves it colourless, not pink, so option A's premise is false. Turmeric stays yellow in water (it does not turn red), and tasting is unsafe and wrong — water is not sweet and the acid is sour, not salty. Only the litmus test reliably tells them apart.

37. (B) Ammonia solution turns red litmus blue, so it is a base. Adding vinegar (an acid) neutralises it, and when neither litmus changes the mixture has reached neutral, where the acid and base have cancelled. It is not still basic, since red litmus would otherwise still turn blue. It is not yet acidic, because that needs excess acid (blue litmus would then turn red). Neutralisation does not create a stronger base.

38. (A) Sour taste and turning blue litmus red are signatures of an acid, and lemon juice is acidic whether concentrated or diluted. Diluting it with water lowers how strongly acidic it is but does not change its family, so both samples still turn blue litmus red. Dilution does

not make an acid stop changing litmus, keep litmus blue, or become neutral; it only weakens the acidity, leaving the acid still an acid.

39. (B) Red litmus turning blue shows washing soda solution is a base. A base turns turmeric red and turns phenolphthalein pink. So turmeric is red and phenolphthalein is pink. Turmeric staying yellow is the acid/neutral result, and phenolphthalein staying colourless is the acid/neutral result, so options pairing those with a base are wrong. Only 'turmeric red and phenolphthalein pink' matches a base on both counts.

40. (B) Acid rain leaves acidic residue on the stone, and a base neutralises an acid, so a dilute basic wash can neutralise that residue and slow further attack on the calcium-carbonate wall. A dilute acid wash would add more acid and worsen the corrosion, so it is unsound. Limestone is a base (calcium carbonate), not acidic, so it is certainly affected by acid rain, and the two washes are not equivalent — only the basic one neutralises the acid.

41. (C) Leaving blue litmus blue (no change) plus turning phenolphthalein pink are both base behaviours; only a base does both. Soap solution is the base, so it fits. Orange juice is acidic (citric acid): it turns blue litmus red and leaves phenolphthalein colourless. Sugar water is neutral: blue litmus stays blue but phenolphthalein stays colourless, so it fails the phenolphthalein test. Hence soap solution is the only match.

42. (B) Lime water is a base. Adding dilute hydrochloric acid neutralises it, and when neither litmus changes the mixture has reached neutral. Adding a few more drops of acid then puts acid in excess, making the mixture slightly acidic. It is not still exactly neutral once extra acid is added; lime water has been used up so the mixture is not basic; and an acid certainly does affect a neutral mixture by making it acidic.

43. (D) Neutralisation always gives a salt and water, but the salt depends on the specific acid and base. Sulphuric acid with potassium hydroxide gives potassium sulphate, not common salt (sodium chloride), because no sodium or chlorine is involved. So labelling it 'common salt' is wrong. No salt-free outcome occurs, the sulphuric acid is not regenerated, and the salt formed is named from this particular acid-base pair: potassium sulphate.

44. (A) Turmeric turns red only in a base and stays yellow in both acids and neutral liquids. So a yellow turmeric result could mean either an acid or a neutral substance, and it cannot separate the two; a second indicator such as blue litmus is needed (it turns red in an acid, unchanged in a neutral). Turmeric does not turn red in acids or neutrals, does not give the same colour in acids and bases, and is not limited to changing only in neutral liquids.

45. (A) Phenolphthalein and methyl orange are the synthetic, lab-made indicators; litmus, turmeric and china rose are natural. So methyl orange (synthetic) paired with china rose (natural) is one of each. Litmus is natural, not synthetic, and china rose is

natural, not synthetic, so the pairings that call litmus or china rose 'synthetic' are wrong, and calling phenolphthalein 'natural' reverses its true kind.

46. (A) Neutralisation of an acid by a base releases heat, so the beaker warms up as the reaction proceeds; nothing is boiling away. Acid boiling away would, if anything, need heat put in and would not be caused by the reaction. A base does not freeze the mixture, and the salt being formed stays dissolved rather than boiling off. The warmth is simply the reaction's released heat.

47. (C) All four choices are bases, but the question asks for the cheap, bulk option actually used to treat acidic water. Slaked lime (calcium hydroxide) is exactly that — inexpensive and spread in large quantities on acidic land and water. Sodium hydroxide (caustic soda) is corrosive and costly, ammonia solution is not used to treat lakes, and milk of magnesia is a medicinal preparation, not a bulk treatment. So slaked lime fits both 'a base' and 'cheap and commonly used'.

48. (A) China rose shows dark pink (magenta) only in an acid. X turns blue litmus red, so X is an acid and turns china rose magenta. Y turns red litmus blue, so Y is a base, and china rose is green (not magenta) in it. Z changes neither litmus, so Z is neutral and leaves china rose its ordinary colour. Hence only X turns china rose magenta; Y, being a base, turns it green instead.

49. (C) Turning blue litmus red and turning methyl orange red are both acid behaviours, and leaving phenolphthalein colourless is consistent with an acid (it is pink only in a base). All the results agree on acidity, so the liquid is an acid. Methyl orange turning red is itself the acid result, not a base sign. Colourless phenolphthalein alone could be neutral, but the two acid results rule that out, so the findings are consistent, not in conflict.

50. (B) China rose is green only in a base, so the solution is a base, and a base turns red litmus blue — that single prediction is safe. The other predictions are acid behaviours: only an acid tastes sour, only an acid turns blue litmus red, and a base turns phenolphthalein pink (not colourless), so 'leaves phenolphthalein colourless' would be wrong for a base. Hence the one correct prediction is that it turns red litmus blue.

51. (D) Brushing the teeth with toothpaste, a mild base, neutralises the acid made by bacteria in the mouth, exactly matching $\text{acid} + \text{base} \rightarrow \text{salt} + \text{water}$. Dissolving sugar is mere mixing, not an acid-base reaction. Ice melting is a physical change of state, not a chemical neutralisation. An apple browning is a reaction with air, not between an acid and a base. Only the toothpaste example is a genuine neutralisation.

52. (C) Exact neutralisation uses up all the acid (the lemon juice) and all the base (the baking soda), leaving a salt dissolved in the water formed. Boiling off the water leaves that solid salt behind, not the original lemon juice — no acid remains after exact neutralisation.

The baking soda is also fully used up, and matter is not destroyed, so a solid does form. The correct answer is the salt.

53. (B) Common salt (sodium chloride) is neutral, not a base, so 'common salt — base' is the wrong pairing. Lemon juice (citric acid) is correctly an acid, vinegar (acetic acid) is correctly an acid, milk of magnesia (magnesium hydroxide) is correctly a base, and soap solution is correctly a base. So the single error is calling neutral common salt a base.

54. (A) Methyl orange is red only in an acid, so a red methyl-orange result agrees with blue litmus turning red and confirms the liquid is an acid. Phenolphthalein turning pink would indicate a base, contradicting the acid finding. Turmeric turning red also indicates a base, again contradicting it. Red litmus turning blue is a base behaviour too. Only methyl orange turning red is a second, independent acid sign.

55. (A) Changing neither litmus paper proves a liquid is neutral, but water is only one of many neutral liquids; salt water, sugar water and similar solutions give the same result. So the test cannot single out pure water, let alone prove it is chemically pure. Water does not turn blue litmus red (it is neutral). Acids and bases do change litmus, so they are not the ones leaving it unchanged. Litmus works fine on colourless liquids, so that objection is false.

56. (D) In acid + base $\rightarrow$ salt + water, the water is genuinely a product of the reaction, formed when the acid and base combine, along with the salt. It is not merely the solvent water that was already present; the equation states water as a product. Leftover hydrochloric acid would be excess acid, not water. And water is always one of the two products, so it is wrong to say only salt forms.

57. (C) Blue litmus turning red is the test for an acid. Vinegar (acetic acid) is an acid, so it fits. Soap solution and milk of magnesia are bases, which turn red litmus blue, not blue litmus red. Distilled water is neutral and changes neither litmus paper. So vinegar is the only acidic liquid among the four that turns blue litmus red.

58. (D) Blue litmus turning red is the test for an acid. Lemon juice and vinegar turn blue litmus red, so two are acids. Soap solution is a base and leaves blue litmus unchanged; lime water is also a base and leaves blue litmus unchanged. So exactly 2 of the four are acids. A count of 1 misses one acid, while 3 or 4 wrongly counts a base as acidic.

59. (B) China rose is magenta in acids and green in bases. Cup P (magenta) is therefore the acid, and cup Q (green) is the base. To neutralise the base in cup Q you add the acid, which is cup P, so adding P slowly to Q neutralises it. Saying magenta marks a base reverses china rose's direction; P is not neutral (it would then merely dilute Q), and china rose certainly distinguishes acids from bases, since it shows two different colours.

60. (B) Phenolphthalein is colourless across the whole neutral range and the whole acidic range, turning pink only once a solution is basic. So at an exact neutral endpoint it gives no

distinctive colour and cannot flag that point precisely. A better confirmation that neither acid nor base is in excess is that both litmus papers stay unchanged — blue stays blue and red stays red. Turmeric red would actually signal excess base, and the other options misstate how phenolphthalein behaves.